Synergistic Activities – Brian Mark Anderson, PhD, DABR, Assistant Professor of Medical Physics, University of California, San Diego

1. **Open-source software for deep learning in radiation oncology.** Author and maintainer of DicomRTTool, a Python library that converts DICOM images, RT structures, and dose grids into NumPy/SimpleITK arrays and converts model predictions back into clinically importable RT structures (Practical Radiation Oncology, 2021); ~1,700 downloads per month, 100+ GitHub stars, 30+ forks, and in use across academic and industry groups internationally. Extended this work into the Keras core library by authoring the reconstruct_patches operation and ReconstructPatches2D/3D layers (merged 2026), which, because Keras 3 is backend-agnostic, reach TensorFlow, PyTorch, and JAX users through a library with ~18 million monthly installs. Also released the DICOM Attribute Manipulation Tool for frame-of-reference and UID editing (Journal of Applied Clinical Medical Physics, 2025).

2. **Nomenclature standardization: AAPM TG-263 and Open RT Structures.** Member of AAPM Task Group 263 (Standardizing Nomenclatures in Radiation Oncology) since 2021 and lead author of Open RT Structures, a freely available implementation of TG-263 that closed the gap between a published nomenclature and one clinics could actually adopt (International Journal of Radiation Oncology, Biology, Physics, 2024). Co-authored the TG-263U1 survey of roadblocks to standardized nomenclature (Journal of Applied Clinical Medical Physics, 2024) and presented the adoption tool at the AAMD Annual Conference (2023). As Technical Service Chief for the central nervous system site at UC San Diego (2025–present), authored the departmental CNS Standards and Guidelines defining site-specific treatment protocols, critical-structure nomenclature, and dose constraints.

3. **Translation of dissertation research into a randomized clinical trial.** Doctoral work at MD Anderson on biomechanical-model deformable image registration and deep-learning segmentation for liver tumor ablation (Society of Interventional Radiology Allied Scientist Grant, $40,000, 2019; Medical Physics, 2021) was translated directly into COVER-ALL (NCT04083378), a Phase 2 randomized controlled trial of software-based versus visual assessment of the minimal ablative margin at the time of the procedure. The trial established software-based assessment as superior to visual assessment (The Lancet Gastroenterology & Hepatology, 2025). The dissertation received MD Anderson's Alfred G. Knudson Jr. Outstanding Dissertation Award (2022).

4. **Hands-on deep-learning training for medical physicists.** Created and taught "getting started with deep learning" workshops that take clinical physicists from DICOM to model predictions, delivered at the Winter Institute of Medical Physics (2020, 2025) and the MD Anderson Image Guided Cancer Therapy Research Program (2020, 2021). Authored open teaching materials for the AAPM Practical Big Data Workshop (GitHub/Jupyter setup and TensorFlow data-pipeline guides, 2021; Early Career Investigator Impact Award) and, as Teaching Assistant for the 2019 AAPM Summer School, helped create and distribute its workbooks. Guest lectured on deep learning at Rice University (ELEC/COMP 576, 2019) and the Georgia Institute of Technology (2023), and as Chief Resident at UC San Diego (2021–2023) coordinated resident education with faculty.

5. **Service to the scientific community and broadening participation.** Associate Editor of Medical Physics and member of its Board of Associate Editors (2023–present; ad hoc Associate Editor 2020–2021). Member of the AAPM Machine Intelligence Subcommittee (2024–present), the AAPM QUANTEC 2 Working Group (2026–present), and Medical Physicists for World Benefit (2017–present). Founding member and domestic liaison of MD Anderson's International Students Association (2019–2021), which helps international graduate students with the transition to the United States; STEM outreach to local students through AAPM WizKids (2018–2019) and UTHealth Science Night (2017–2019); and Co-Investigator on AHRQ R18 HS029474 (2024–2025), developing AI-enhanced pre-treatment peer review to improve patient safety in radiation oncology.